A County-Level Look at Quality-of-Life Tradeoffs Across the U.S.

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Section 1. Introduction

I chose this topic because I am thinking about where I might want to make a long-term home. This project compares selected quality-of-life measures across U.S. locations, including air quality, cost of living, income, unemployment, and crime. The measures mostly contain data from 2022, unless notes elsewhere and county level data.

This project is a descriptive comparison.

https://public.tableau.com/app/profile/alex.heyert/viz/Alex_Heyert_final/USAStory

Section 2. Data Description

Domain

The data describes measurable conditions related to quality of life across the United States. It includes economic, environmental, population, and public-safety measures.

Data file and project fields

- The Kaggle source dataset contains 3,134 rows and 34 columns. For this project, I used a cleaned analysis table containing 3134 rows and 12 columns: 6 measure fields and three geographic identifiers: FIPS code, LSTATE (State name), and MNCNTY(County name). Several other columns were kept but not used such as LZIP, and LOCALE and more detailed income measures.

Data source and measures

The project dataset was downloaded from

Kaggle: City/ZIP/County/FIPS – Quality of Life.

The Kaggle dataset combines information from

several public sources. The measures used in this project

Were curated by the author from the sources shown

on the right.

Measure used	Source	Time Period
Air-quality measure: share of days rated good	U.S. EPA Outdoor Air Quality Data	2022
Population	U.S. Census Bureau county population estimates	2022
Median income and cost of living	Economic Policy Institute family budget data	2022
Unemployment	USDA Economic Research Service county data	2022
Reported crime rate	ICPSR 2016 reported-crime data	2016

Section 3. Data Cleaning Strategies

I used Excel to prepare the analysis table and Tableau to create a numeric crime-rate measure.

- The value -1 indicates insufficient data. I replaced it with a blank for the applicable measure fields so it would be treated as missing, not as a real measurement or zero.
- I removed the incomplete fields Overall rank, ParkScore2023 Rank, and %CvgCityPark.
- The crime-rate field was stored as text in a format such as 9/1000. I created a Tableau calculated field to extract the numeric rate, reported as crimes per 1,000 residents.

Section 4. Clean Dataset

The clean dataset used for this project contains **3,134 rows and 12 columns**. Each row represents a county-level record. The table includes six measure fields—air quality, population, median income, cost of living, unemployment, and crime rate—and the location fields FIPS code, state, county, ZIP code, and location type. ZIP code and location type were retained in the dataset but were not used in the visualizations.

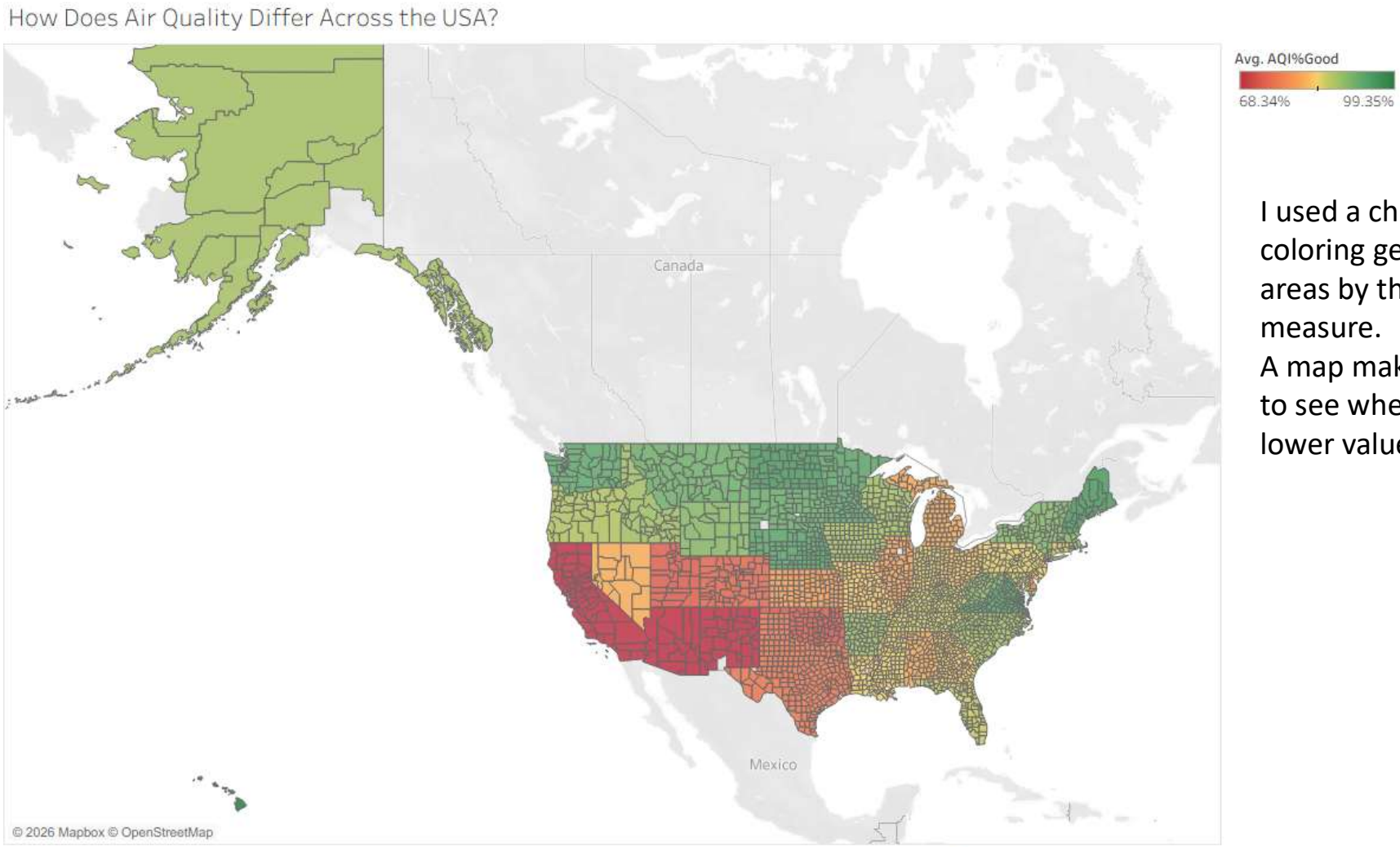
	A	B	C	D	E	F	G	H	I	J	K	L
1	countyhelper	LSTATE	NMCNTY	FIPS	LZIP	ULOCALE	2022 Population	2016 Crime Rate	Unemployment	AQI%Good	Cost of Living	2022 Median Income
2	VACharles City County	VA	Charles City County	51036	23030	42-Rural: Distant	6,605	8/1000	3.21%	93.76%	$75,531.37	$78,038.78
3	TXMcmullen County	TX	McMullen County	48311	78072	43-Rural: Remote	576	47/1000	1.81%	75.33%	$63,913.28	$67,513.81
4	TXTerrell County	TX	Terrell County	48443	79848	43-Rural: Remote	693	20/1000	3.54%	75.33%	$64,361.02	$55,946.62
5	AKSkagway Municipality	AK	Skagway Municipality	02230	99840	43-Rural: Remote	1,081	13/1000	7.19%	87.86%	$87,709.32	$85,446.30
6	GABaker County	GA	Baker County	13007	39870	42-Rural: Distant	2,788	0	4.19%	83.30%	$59,389.29	$52,946.23
7	KSGeary County	KS	Geary County	20061	66641	41-Rural: Fringe	35,691	23/1000	3.91%	79.67%	$69,401.74	$59,946.11
8	COHinsdale County	CO	Hinsdale County	08053	81235	43-Rural: Remote	775	13/1000	2.58%	73.82%	$73,190.49	$66,626.63
9	TXCochran County	TX	Cochran County	48079	79379	43-Rural: Remote	2,526	33/1000	5.08%	75.33%	$61,721.15	$59,858.82
10	TXArmstrong County	TX	Armstrong County	48011	79019	42-Rural: Distant	1,850	7/1000	2.85%	75.33%	$73,725.97	$81,611.11
11	TXDelta County	TX	Delta County	48119	75469	43-Rural: Remote	5,406	6/1000	4.18%	75.33%	$62,084.03	$63,549.24
12	TXBorden County	TX	Borden County	48033	79738	43-Rural: Remote	585	19/1000	1.89%	75.33%	$64,935.10	$98,648.20
13	NDDivide County	ND	Divide County	38023	58730	43-Rural: Remote	2,187	1/1000	1.34%	95.24%	$69,986.84	$91,452.21
14	OKHarmon County	OK	Harmon County	40057	73550	43-Rural: Remote	2,428	40/1000	2.82%	72.44%	$69,744.14	$62,484.42
15	NCTyrrell County	NC	Tyrrell County	37177	27925	43-Rural: Remote	3,365	14/1000	5.47%	88.12%	$72,056.78	$51,896.82
16	VAMathews County	VA	Mathews County	51115	23109	42-Rural: Distant	8,490	9/1000	2.94%	93.76%	$77,641.60	$68,953.42
17	ORSherman County	OR	Sherman County	41055	97039	43-Rural: Remote	1,955	5/1000	3.44%	86.98%	$82,162.30	$64,759.86
18	SDJones County	SD	Jones County	46075	57559	43-Rural: Remote	884	0	2.55%	92.09%	$74,091.73	$53,908.37
19	VAWilliamsburg City	VA	Williamsburg city	51830	23185	23-Suburb: Small	15,909	25/1000	4.26%	93.76%	$82,897.39	$88,563.75
20	VANelson County	VA	Nelson County	51125	22920	42-Rural: Distant	14,652	14/1000	2.85%	93.76%	$74,618.57	$80,136.59

Section 5. Visualization Tools

- I used Excel to clean and prepare the data. I used Tableau to create the charts, geographic views, dashboards, and stories. Tableau was useful for mapping the geographic fields and comparing measures across locations.
- **Tableau Public project:** **https://public.tableau.com/app/profile/alex.heyert/viz/Alex_Heyert_final/USAStory**

Section 6. Visualizations and Stories

Goal:

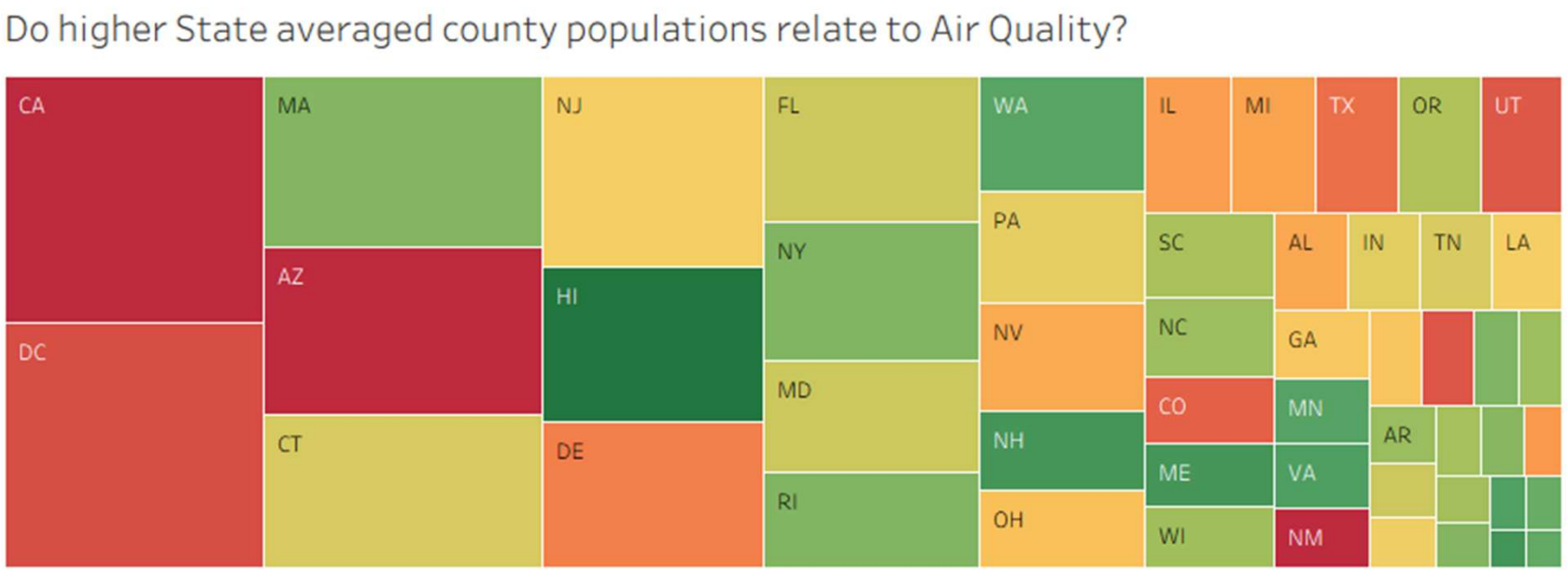


I used a choropleth map, coloring geographic areas by the air-quality measure.
A map makes it possible to see where higher and lower values are located.

Story:

Air quality appears to be the worst in the southwest. We also see poor air quality in parts of the midwest and larger states like Florida and pennsylvania. Northern and non-continental states show better air quality. Could air quality be linked to population?

Goal:

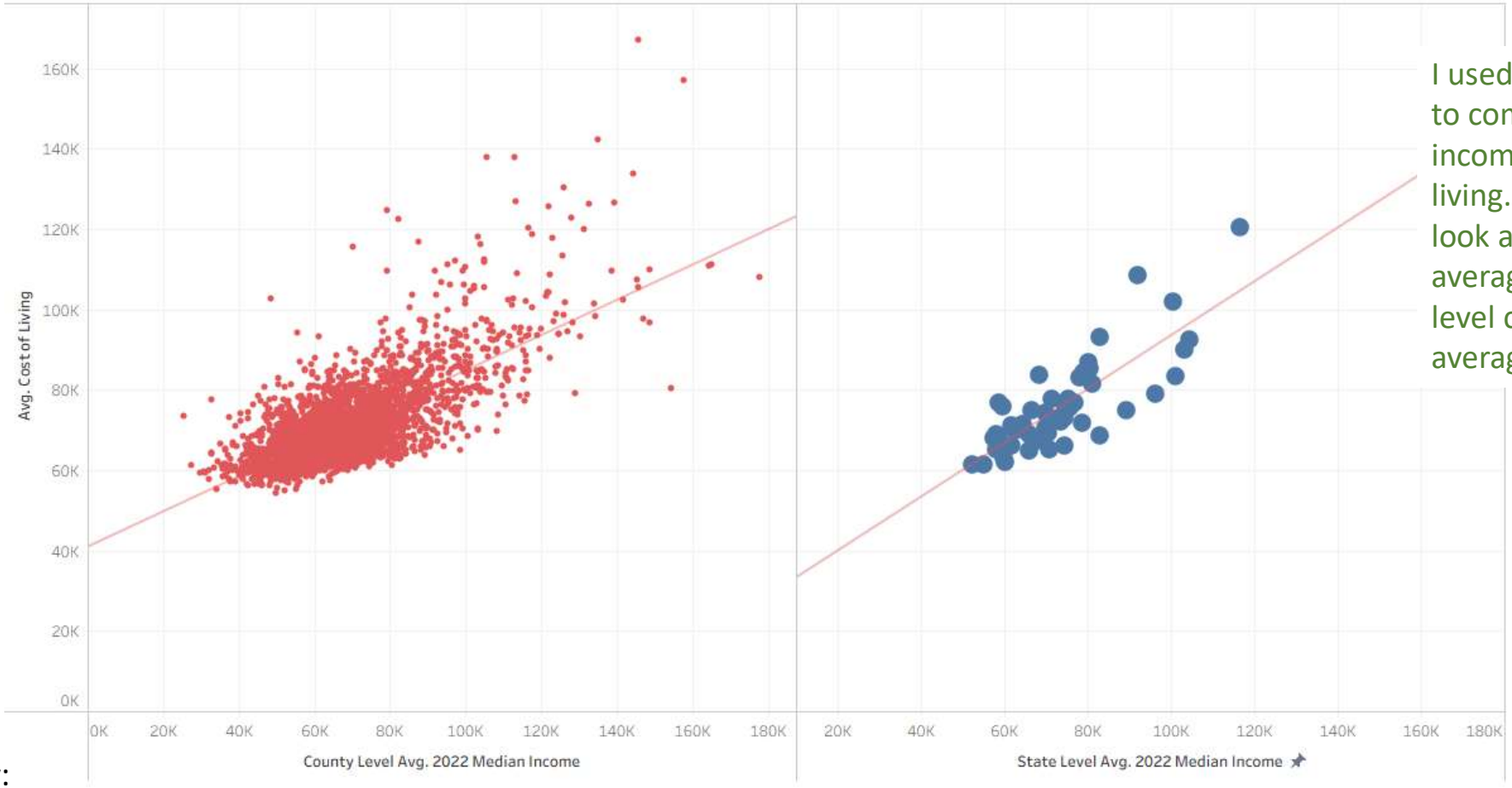


Story:

Larger Population appears to be linked with poor air quality, with a few exception such as New Mexico, Utah, and Oklahoma

I used a treemap with tile size representing average county population across each state and tile color representing the state's county level average of good air quality days. This lets viewers compare population and air quality in the same view.

Goal: Does Income Scale with the Cost of Living?



I used a scatterplot to compare median income with cost of living. I chose to look at county averages and state level county averages.

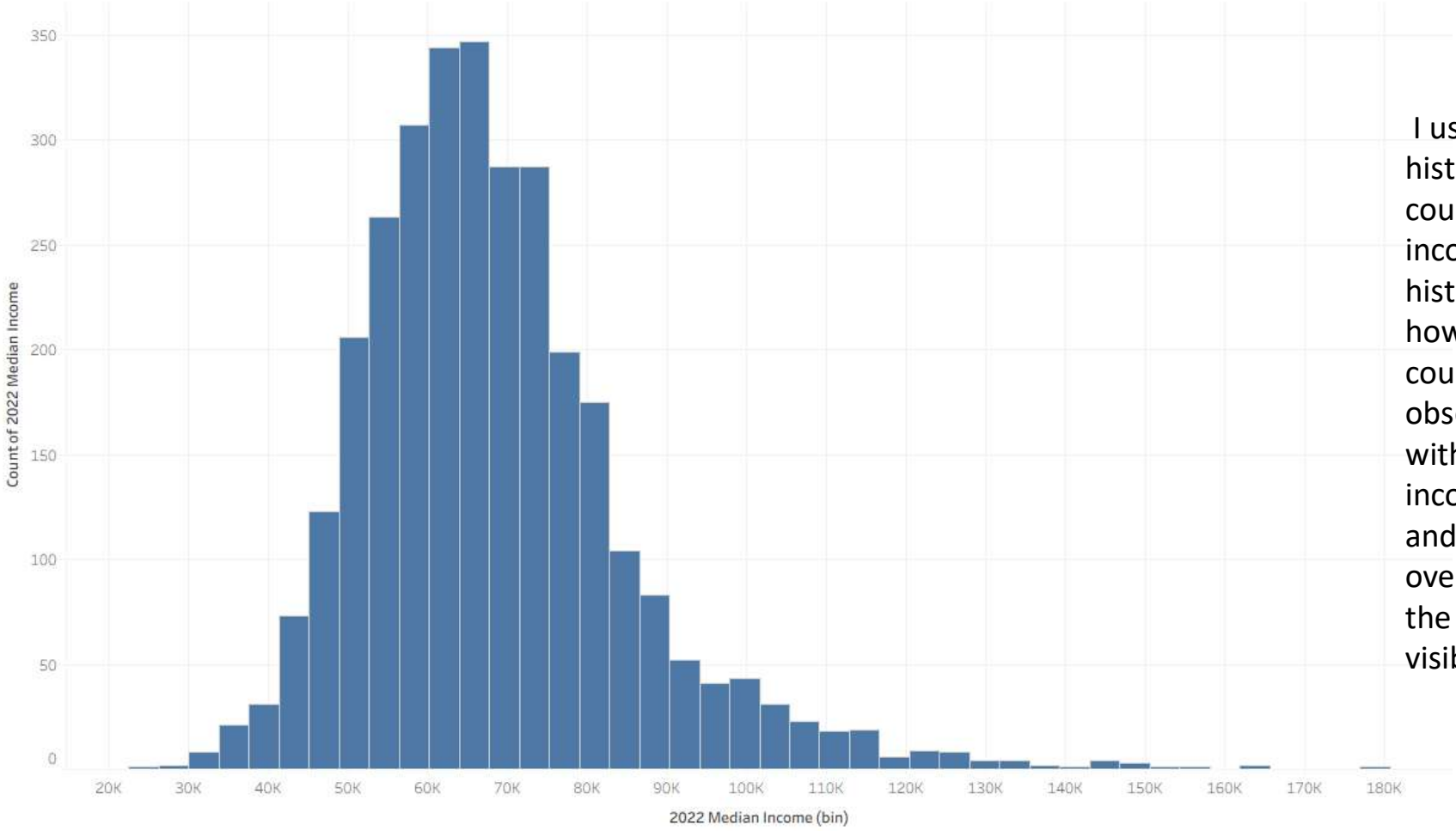
Story:

This question was asked with two levels of detail - county averages and state averages.

On a state level there isn't a perfect scale, but income does seem to increase with the cost of living.

On the county leve the data is much more dispersed.

Goal: How are county median incomes distributed?

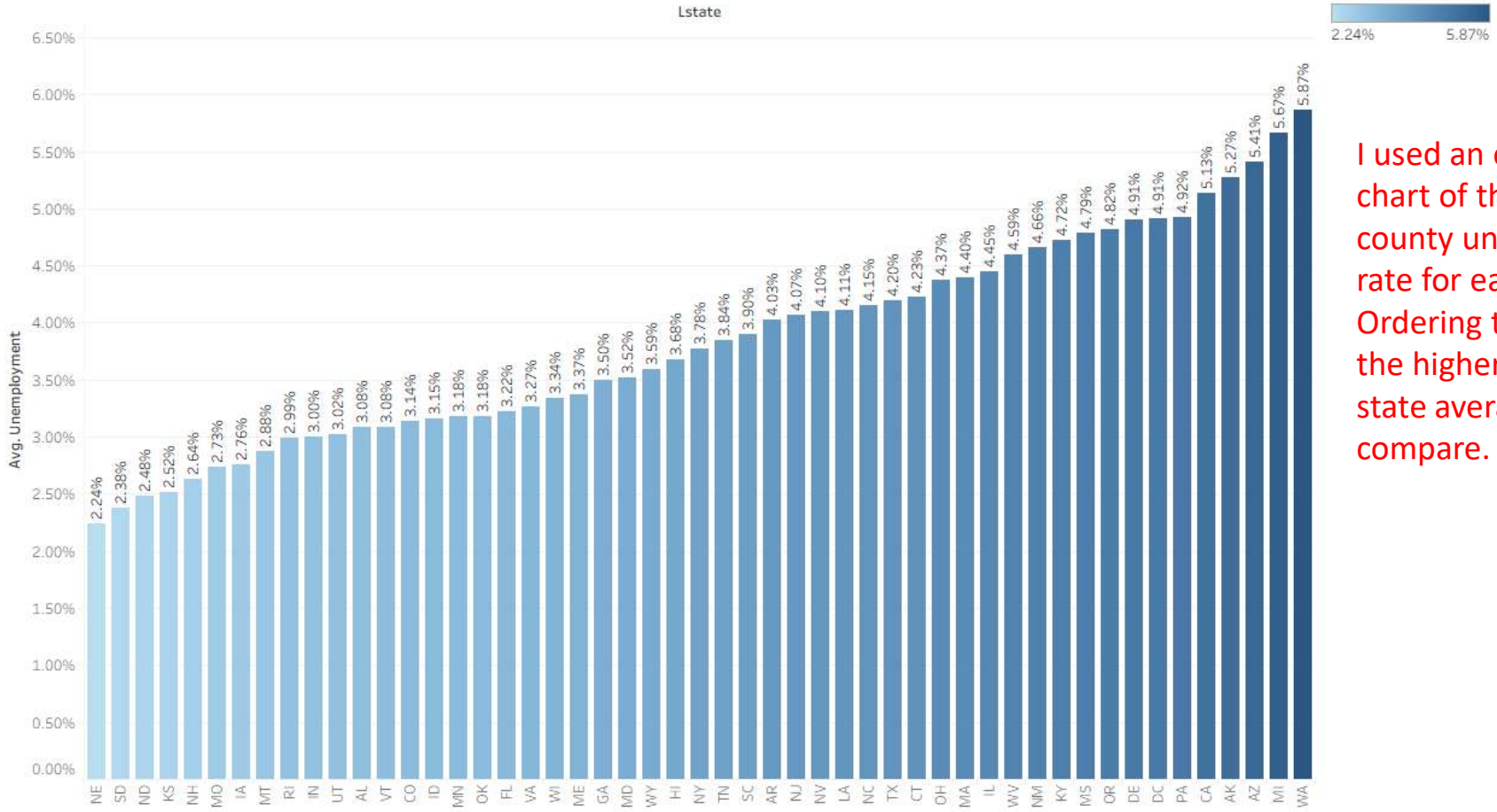


I used a histogram of county median income. A histogram shows how many county observations fall within each income range and makes the overall shape of the distribution visible.

Story:

At first glance it looks like income is evenly distributed, most counties earn 50-80k annually. The data does show a right skew.

Goal: Which states have the most unemployment?

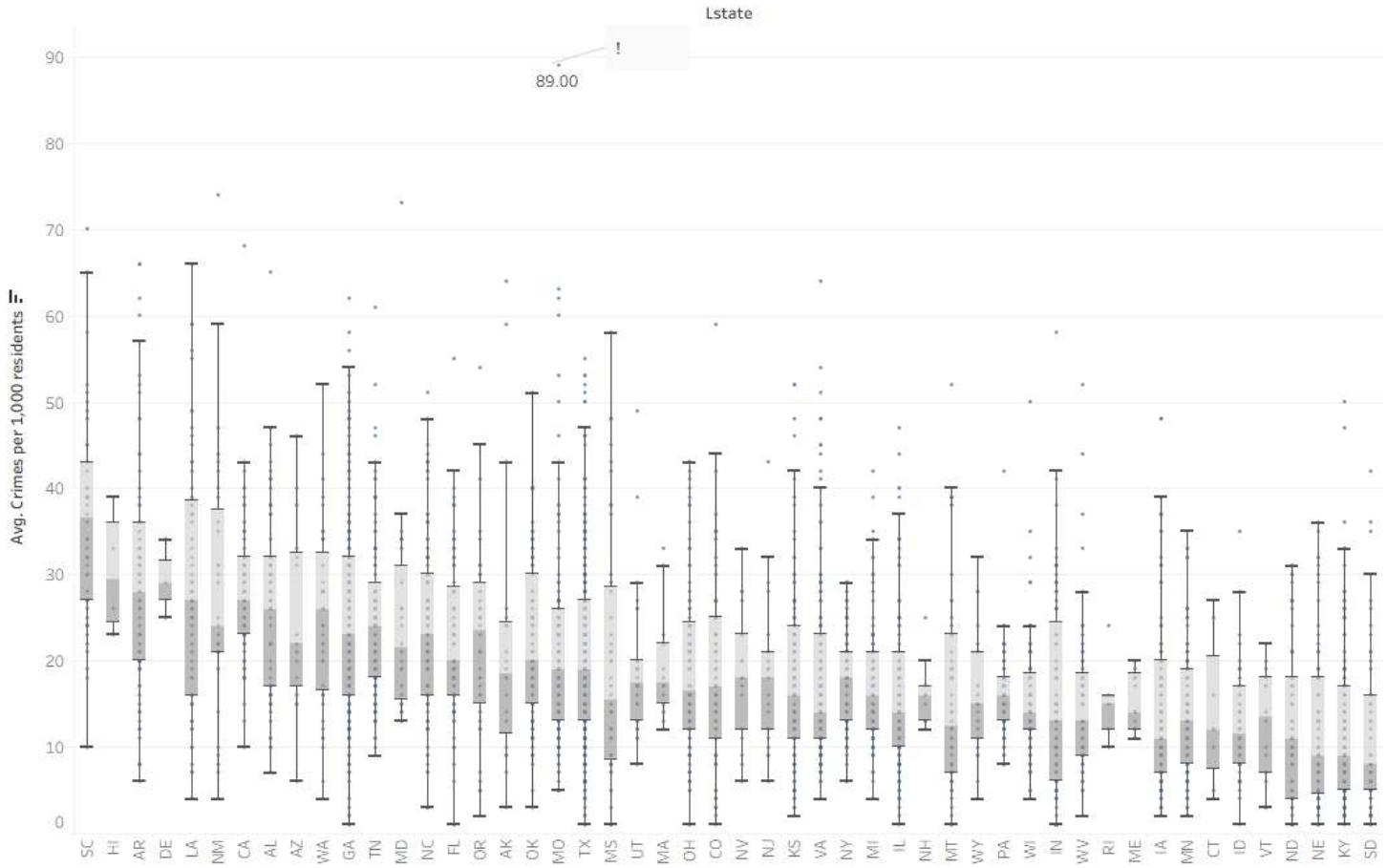


I used an ordered bar chart of the average county unemployment rate for each state. Ordering the bars makes the higher and lower state averages easier to compare.

Story: Washington, Michigan, and Arizona had the highest unemployment rates in 2022, but nearly half of the states had a relatively high unemployment rate, above 3.5%.

Goal:

How do crime rates vary across the states and cities?



I used a box-and-whisker plot of 2016 reported crime rates per 1,000 residents, grouped by state. A box plot shows both the typical range and unusual city level values within each state.

Story:

Crime rates vary across states, and many states also show substantial variation among their counties. St. Louis is the most prominent outlier in this view, at 89 reported crimes per 1,000 residents. This data is from 2016 and may not reflect current crime patterns.

Section 7. Conclusions

- I've lived in Massachusetts, Florida, Iowa, Arizona, Missouri, Nevada, and California, so I wanted to use data to compare what different places are like. The charts show that quality of life can vary a lot, and no one measure tells the whole story. I've lived in Massachusetts, Florida, Iowa, Arizona, Missouri, Nevada, and California, so this project felt personal to me. I already knew I didn't want to return to Arizona. But the 2022 air quality results for Missouri might have made me think twice about moving to St. Louis or at least look more closely before deciding.
- Iowa didn't stand out at either end on most of the measures, and that fits my experience: it was an affordable place for me to work. The unemployment chart showed a bigger difference between states: Washington had the highest average county rate at 5.87%, while Nebraska had the lowest at 2.24%. Most county median incomes were between $50,000 and $80,000, and higher-income counties also tended to have higher costs of living.
- The crime data is from 2016, and St. Louis stood out at 89 reported crimes per 1,000 residents. That result is interesting, but it may not describe current conditions. The charts don't tell me where I should live, but they show which places or issues may warrant a closer look.
- These charts can't dictate the best place to live, but they could help rule out places based on what factors are important.